Baiyu Peng

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EDUCATION

EPFL (Swiss Federal Technology Institute of Lausanne) **Lausanne, Switzerland**

Ph.D. student in Robotics, Control, and Intelligent Systems 09/2022 - present

- **Ph.D. Thesis: Learning Explicit Constraints from Expert Demonstrations**

Tsinghua University **Beijing, China**

Master in Mechanical Engineering 08/2019- 09/2022

- **Academic**: GPA: 3.64/4.0
- **Master's Thesis:** Chance Constrained Reinforcement Learning for Dynamic Obstacle Avoidance of Intelligent Mobile Robots | **Winner of Master Thesis Award of Tsinghua University**

Tsinghua University **Beijing, China**

Bachelor in Vehicle Engineering and Economics (Minor) 08/2015- 07/2019

- **Academic**: GPA: 3.74/4.0
- **Awards**: National Scholarship, 3/75 Top 3%, 2017 | Outstanding Graduates of Beijing, 5/75 Top 5%, 2019 | Excellent Graduates of Tsinghua University, Top 10%, 2019 | Comprehensive Excellence Scholarship, Top 10%, 2018
- **Bachelor's Thesis Project**: End-to-end autonomous driving though deep reinforcement learning

PUBLICATION

Journal

- **Baiyu Peng,** Albert Árboles, Aude Billard. "Learning Equality Constraints from Trajectory Demonstrations." IEEE Robotics and Automation Letters, 2026. (accepted)
- **Baiyu Peng**, Aude Billard. "Actively Learning Task Constraints from Feasible and Failed Demonstrations." IEEE Transactions on Artificial Intelligence, 2026. (accepted)
- **Baiyu Peng**, Aude Billard. "Learning Stage-wise Constraints from Expert Demonstrations." 2026 (underr eview)
- **Baiyu Peng**, Aude Billard. "Positive-Unlabeled Constraint Learning for Inferring Nonlinear Continuous Constraint Functions from Expert Demonstrations," IEEE Robotics and Automation Letters, 2024.
- **Baiyu Peng**, Jingliang Duan, Jianyu Chen, et al. "Model-based Chance-Constrained Reinforcement Learning via Separated Proportional-Integral Lagrangian." IEEE Transactions on Neural Networks and Learning Systems, 2022.
- **Baiyu Peng**, Qi Sun, Shengbo Eben Li, et al. "End-to-End Autonomous Driving through Dueling Double Deep Q-Network." Automotive Innovation 4(3), 328–337, 2021.

Conference Proceedings

- **Baiyu Peng**, Yao Mu, Jingliang Duan, et al. "Separated Proportional-Integral Lagrangian for Chance Constrained Reinforcement Learning." 32nd IEEE Intelligent Vehicle Symposium (IV), 2021. (**Student Best Paper Award Finalist, Top 1%**, 3/220, [video])
- **Baiyu Peng**, Yao Mu, Yang Guan, et al. "Model-Based Actor-Critic with Chance Constraint for Stochastic System." 60th IEEE Conference on Decision and Control (CDC), 2021. ([video])
- Mu, Yao, **Baiyu Peng**, Ziqing Gu, et al. "Mixed Reinforcement Learning for Efficient Policy Optimization in Stochastic Environments." 20th International Conference on Control, Automation and Systems (ICCAS), 2020. (**Student Best Paper Award, Top 1%**, 5/500)

RESEARCH EXPERIENCE

My research interests and experiences mainly include reinforcement learning (RL) and its application in automated vehicles and robots. I especially pay attention to (1) improving RL learning efficiency by using the model of the vehicle/robots to optimize policy (2) equipping RL with chance safety constraints for real-world safety-critical scenarios under uncertainty.

In addition to RL, I also have extensive knowledge and experience in classical control methods such as Model Predictive Control (MPC).

Safe Risk-Averse Reinforcement Learning | Project Leader 04/2020-09/2022
Intelligent Driving Lab, Tsinghua University

- Proposed two model-based risk-averse RL algorithms that learns a control policy with a high probability of being safe under model uncertainty. The proposed methods have less oscillations and conservatism than baselines, with a fast learning process.
- Conducted a real mobile robot obstacle-avoidance experiment. The trained robot was able to reach the destination without colliding with a randomly moving obstacle. (video)
- Published and orally presented two conference papers as the first author, one of which won Student Best Paper Award Finalist.

Multi-Robot Distributed Planning and Control | Project Leader 10/2020-09/2022
Geek+ Robotics Tech. Co. Ltd. & Intelligent Driving Lab, Tsinghua University

- Developed an integrated decision-making and control framework for mobile robots, which consists of an A-star static path planner and an MPC dynamic optimal tracker.
- Developed the communication module, path planner module and controller module of the mobile robots. Conducted extensive real-world experiments, where the two robots reached their own destinations without colliding.
- See video for a quick overview and the experimental performance.

Model-based Reinforcement Learning | Main Participant 10/2019-04/2020
Intelligent Driving Lab, Tsinghua University

- Derived the Bayesian estimator to update the stochastic model by collected data.
- Designed and accomplished an aircraft system simulation to verify the optimality of the proposed method.
- Published a conference paper as the second author and won Student Best Paper Award.

End-to-end Autonomous Driving via Reinforcement Learning | Project Leader 11/2018-06/2019
Intelligent Driving Lab, Tsinghua University

- Developed an RL-based end-to-end driving method and verified it on a car simulator. The proposed method used a dual network with both camera image and motion information as the inputs to improve the performance.
- Drew saliency maps of the driving policy network via gradient-based ConvNet visualization technique. Analyzed the mechanism of the RL agent based on the saliency maps.
- Published a journal paper as the first author.

Robust-Control-Based RL Driving Policy Transfer | Main Participant 07/2018-09/2018
Mechanical Systems Control Lab, UC Berkeley

- Accomplished a car trajectory tracking simulation to verify the proposed RL method.
- Wrote and deployed the ROS code in the experimental car. Conducted a real autonomous driving experiment, where the RL controller trained in the simulator drove the real car robustly along a 300 m test road without getting out of road.

TEACHING EXPERIENCE

Teaching Assistant 02/2023-07/2026
EPFL

- Applied Machine Learning (Fall 2023, Fall 2024)
- Machine Learning I (Fall 2025)
- Robotics Practicals (Spring 2023, Spring 2024, Spring 2026)

STUDENT SUPERVISION

EPFL

Master Thesis 05/2026 - present

- Pengcheng Ren, Bayesian Constraint Learning from Demonstration and Language Guidance

Bachelor Thesis 09/2025-01/2026

- Albert Árboles, Learning Equality Constraints from Demonstrations

Master Semester Project 02/2026-06/2026

- Vincent Bouzereau, Model Predictive Control for Constraint Learning from Expert Demonstrations
- Marcus Edjolo, LLM-guided Constraint Learning from Demonstration

Master Research Assistant 03/2023-06/2023

- Erfan Etesami, Constraint Learning via Dynamical Systems Approaches

DUTIES & ACTIVITIES

Workshop Lecturer (Volunteer) 09/2018-02/2022

Center for Student Studying and Development, Tsinghua University

- Organize regular workshops and give lectures to new students on the topics of academic and career planning. (9 workshops, 1 video, served more than 350 students)

Vice Minister of Publicity Department 09/2017-01/2018

The University Student Union, Tsinghua University

- Organize department members to make social media posts about the campus life and other useful information. (10 posts, 50000 reads)
- Publish and propagate the activity information by social media for other departments of the University Student Union.

SKILLS

Programming: Python, Pytorch, ROS, Matlab, C++; Git, LaTex

English: TOEFL 102 (Reading 30 | Listening 28 | Speaking 21 | Writing 23)